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How motivational feedback increases user's benefits and continued use: A study on gamification, quantified-self and social networking

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Highlights

- We study effect of gamification, quantified-self and social networking features.
- Mediating variables are affective, informational and social feedback.
- Dependent variables are perceived benefits and continued use intention.
- Context of the study is a mobile motivational information system for health.
- Overall, motivational design increases feedback, benefits and continued use.

Abstract

With the increasing provenance of hedonic and social information systems, systems are observed to employ other forms of feedback and design than purely informational in order to increase user engagement and motivation. Three principle classes of motivational design pursuing user engagement have become increasingly established; gamification, *quantified-self* and *social networking*. This study investigates how the perceived prominence of these three design classes in users' use of information system facilitate experiences of *affective*, *informational* and *social feedback* as well as *user's perceived benefits* from a system and *their continued use intentions*. We employ survey data (N = 167) gathered from users of HeiaHeia; an exercise encouragement system that employs features belonging to the three design classes. The results indicate that gamification is positively associated with experiences of affective feedback, quantified-self with experiences of both affective and informational feedback and social networking with experiences of social feedback. Experiences of affective feedback are further strongly associated with user perceived benefits and continued use intentions, whereas experiences of informational feedback are only associated with continued use intentions. Experiences of social feedback had no significant relationship with neither. The findings provide practical insights into how systems can be designed to facilitate different types of feedback that increases users' engagement, benefits and intentions to continue the use of a system.

Introduction

Since their inception, information systems have mostly been designed with instrumental ends in mind (Davis, 1989; Hirschheim & Klein, 2012; McLeod & Schell, 2008). In other words, information systems have thus far mainly attempted to make people more efficient through better provision of information and informational feedback. As technological developments took place, systems, services and their facilitating infrastructure became more varied. The focus of information systems departed from the mindset of efficiency and instrumentality towards facilitating human life in manifold ways such as through entertainment (e.g. games) and social connections (e.g. social networking applications) (Hamari, 2015; Hassan, 2018; Oinas-Kukkonen, 2012). We have since observed an expansive understanding of user benefits and the feedback systems can provide their users to include not only informational benefit/feedback (e.g. performance reports) (Davis, Bagozzi, & Warshaw, 1989) but also affective benefit/feedback (e.g. entertainment, enjoyment) (van der Heijden, 2004) and social benefit/feedback (e.g. social interaction) (Boyd & Ellison, 2007) amongst others (Giboney, Briggs, & Nunamaker, 2017).

Today, it appears that the majority of contemporary systems and services attempt to facilitate user engagement, not only through providing users accurate information but additionally through providing them affective and social experiences to further support both the utility and the self-purposefulness of system use (Fang, Zhao, Wen, & Wang, 2017; Giboney et al., 2017; Oinas-Kukkonen & Harjumaa, 2018; Zhang, 2008). However, even beyond both utility through efficiency and self-purposefulness through enjoyment, today's systems attempt to merge both notions and employ hedonic design to spur increased utility through increasing positive user motivation. Therefore, modern information systems can be increasingly considered "motivational information systems" (motivational ISs) (Hamari, 2015; Koivisto & Hamari, 2019). Whereas utilitarian information systems primarily aim at efficiency and productivity and whereas hedonic information systems (such as games) primarily aim to afford self-purposeful system use, motivational ISs aim to increase motivational affordance by implementing hedonic systems design, primarily aiming to increase productivity through increased motivation and engagement (Hamari, 2015; Hamari, Hassan, & Dias, 2018; Koivisto & Hamari, 2019).

For example, in the realm of exercise: motivational ISs such as exergames, motivate users towards exercising more through for example affective feedback, social connections, social comparison, next to traditional informational feedback and visual summaries of performance (Huang, Pham et al., 2018; Nguyen et al., 2018). Similar motivational systems have been introduced to for example: educational contexts (Coleman, 2018), and organizational contexts (Jung, Schneider, & Valacich, 2010).

At the helm of the motivational IS's development has notably been three popular conceptual and technological design classes (Hamari et al., 2018): 1) Gamification - drawing design from games with the aim of making activities, services and systems more intrinsically motivating and enjoyable by providing users affective feedback (Huotari & Hamari, 2017; Landers, Auer, Collmus, & Armstrong, 2018), 2) Quantified-self (QS) - benefiting from advances in big data and wearables, QS designs collect, store and visualize data related to users' performance, attempting to provide users personal and precise informational feedback (Lupton, 2016; Nafus & Sherman, 2014) and 3) Social networking - drawing from the designs of social networking services and social computing to attempt to invoke socio-psychological responses, such as a sense of community through social feedback (Boyd & Ellison, 2007; Chen, Lu, Chau, & Gupta, 2014). As discussed, these design developments are prominently observed in consumer applications and systems in various areas such as in exercise (Fitocracy, Myfitnesspal), education (Codecademy, Khan Academy) and Habits formation (Habitica, Superbetter). The popularity of the three classes of design is also observed in how established system packages have come to employ gamification and social features as seen in many ERP implementations (e.g., Alcivar & Abad, 2016; Herzig, Strahringer, & Ameling, 2012).

Despite the growing prominence of gamification, quantified-self and social designs both in academia and industry, we lack a concrete and comparative understanding of how these designs provide the informational, affective and social feedback that they are assumed to provide to engage and benefit users. The veins of literature on the respective three design classes and the feedback they could be providing remain, for the most part, separate and distinct as each design practices are separately investigated by respective veins of literature. While features from the three design classes may be simultaneously present in a motivational IS, the features are likely to overshadow one another, and users are likely to view features differently, being more attuned to the features that they find prominent. There is, hence, a need to determine what feedback do users experience from the features they attune to the most and if the feedback users experience from these designs does lead to perceived benefits from, or intentions to continue the use of a system or a service that employs these designs. It is hence pertinent theoretical, as well as a practical, questions to ask:

- 1) How does the prominence of gamification, quantified-self and social networking features in users' use of a system affect experiences of affective, social and informational feedback?

- 2) How does experiences of affective, social and informational feedback affect the perceived beneficial-ness of the system use and their continued use intentions?

Investigating these questions assists 1) designers in designing motivational systems and services that employ features and provide feedback that best resonates with users, and 2) individuals and organizational managers in selecting technologies that provide individuals and organizations benefits and that are used more longitudinally, which could help in attaining more returns from IS investments. To answer the outlined research questions, we utilized data from a survey (N = 167) of users of HeiaHeia; an exercise encouragement system which employs features belonging to the three design classes: gamification, quantified-self and social networking allowing for the simultaneously study of the feedback these designs provide.

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Section snippets

Feedback and Motivational information system design

The, affective and social responses and performance information people attain in response to their behavior are what is commonly referred to as feedback (Carver & Scheier, 1990; Hattie & Timperley, 2007; op den Akker, Jones, & Hermens, 2014). One of the main purpose of feedback is to help individuals reduce the discrepancies between intended and actual behavior (Hattie & Timperley, 2007; Shute, 2008). Hence, it can guide an individual's attention towards their intended performance (Fishbach & ...

Participants

Data (N = 167) was gathered via an online survey of users of HeiaHeia: an exercise tracking system that incorporates features of gamification, QS and social networking. Gamification features included a leaderboard of the sports a user played the most, medals earned, and levels reached. Social networking features included cheering, commenting, viewing of

friends' activity logs on a timeline and a list of the friends a user interacts the most with. Quantified-self features included advanced ...

Results

The investigated perceived prominence of the three classes of motivational design: gamification, quantified-self and social networking) accounted for 24.5% of the variance in the experiences of informational feedback, 70.6% of the variance in the experiences of social feedback and 30.4% of the variance in the experiences of affective feedback. Perceptions of experiencing affective, social and informational feedbacks account for 12.5% of the variance in the perceived benefits from the use of the ...

Discussion

The results obtained by this study indicate that the 1) experience of affective feedback is facilitated by the perceived prominence of gamification and quantified-self designs and is positively associated with intentions to continue the use of a motivational IS and is solely and positively associated with perceived benefits from the use of a motivational IS, 2) experience of social feedback is facilitated by the perceived prominence of social networking designs, negatively associated with the ...

Conclusion

This paper investigated the three main design classes of motivational information systems, namely: gamification, quantified-self and social networking features and their effects on user experience. As our results indicate, users of these design classes experience a combination of different feedback types ranging between affective, social and informational feedback. Out of these feedback types facilitated by motivational systems design classes, it appears that affective feedback, often ...

Declaration of interest

The authors declare not conflicting interest whatsoever with the journal. ...

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...Badges upgrading is positively related to perceived enjoyment. Effective gamification mechanism design could remarkably drive groups of people together in the form of collaboration and socializing (Hamari & Koivisto, 2013; Hassan *et al.*, 2019). This is beneficial to enhance social interaction among individuals....

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